Raffles Institution
Year 5-6 Physics Department

2024 Year 6 H2 Physics Term 3 Timed Practice Solution / Mark Scheme

Section B

1 **(a)** When the path difference of the waves from each slit to the same point on the screen is integer multiples of the wavelength of the light, the waves meet in phase and there is constructive interference, forming a fringe with maximum intensity. B1

When the path difference of the waves from each slit to the same point on the screen is odd integer multiples of half of the wavelength of the light, the waves meet π rad out of phase and there is destructive interference, forming a fringe with minimum intensity. B1

(b) double-slit interference pattern,
$a\sin\theta = n\lambda$ ----- (1) M1
first minimum of single-slit diffraction pattern,
$b\sin\theta = \lambda$ ----- (2)

$$\frac{(1)}{(2)} \qquad n = \frac{a}{b} = \frac{0.60}{0.10} = 6$$

The sixth orders from the interference pattern coincide with the first minima of the single-slit diffraction pattern and are missing.

The width of the central fringe is the distance between the two first minima of the single-slit diffraction pattern on either side of P, which is twice the distance between the $n = 0$ and $n = 6$ maxima of the double-slit interference pattern.

From Fig. 1.2, fringe separation $x = 11.6 - 10.0 = 1.6$ mm
width of central fringe $= 2(6x) = 12(1.6) = 19.2$ mm A1

(c) **(i)** The contrast decreases as the intensity of the maxima decreases. B1

$$x = \frac{\lambda D}{a}$$
The fringe separation remains the same. B1

$$n' = \frac{a}{b'} = \frac{a}{0.5b} = 2 \times \frac{a}{b} = 2 \times 6 = 12$$
The twelfth orders are missing from the interference pattern. B1
OR
The width of the central fringe of the single-slit diffraction pattern is doubled.

(ii) The contrast remains the same. B1

$$x' = \frac{\lambda' D}{a} = \frac{2\lambda D}{a} = 2x$$
The fringe separation is doubled. B1

The missing orders from the interference pattern remains unchanged at $n = 6$. B1
OR
The width of the central fringe of the single-slit diffraction pattern is doubled.

2 **(a)** When the molecules of a gas collide into the walls of the container, they experience a change in momentum, and hence a normal force by the container wall. By Newton's third law, the molecules exert an equal and opposite force on the container wall. B1

The force exerted per unit surface area of the container walls by all the molecules, gives rise to pressure. B1

(b) **(i)** The straight-line graph through the origin implies that density is directly proportional to the pressure of the gas at a constant temperature, $\rho \propto p$. M1

For a fixed mass of gas, $\rho = \frac{M}{V}$. Hence $\rho \propto \frac{1}{V}$ and $p \propto \frac{1}{V}$.

From the ideal gas equation $pV = nRT$, when temperature T is constant, A1

$p \propto \frac{1}{V}$. Hence the gas behaves like an ideal gas.

(ii) pressure of a gas is given by the equation (from formula sheet):

$$p = \frac{1}{3}\frac{Nm}{V}\langle c^2 \rangle$$

since $\rho = \frac{Nm}{V}$, $p = \frac{1}{3}\rho\langle c^2 \rangle$

equation of the graph is $\rho = \frac{3}{\langle c^2 \rangle} p$ M1

gradient of the $\rho - p$ graph $= \frac{3}{\langle c^2 \rangle}$

at $T = 290$ K, using points $(0.000\ ,\ 0.00)$ and $(2.000 \times 10^5\ ,\ 2.50)$,

$$\text{gradient} = \frac{2.50 - 0.00}{(2.000 - 0.000) \times 10^5} = \frac{2.50}{2.000 \times 10^5} = 1.25 \times 10^{-5}$$ M1

$$\frac{3}{\langle c^2 \rangle} = 1.25 \times 10^{-5}$$

$$c_{rms} = \sqrt{\langle c^2 \rangle}$$

$$= \sqrt{\frac{3}{1.25 \times 10^{-5}}}$$

$$= 489.90 = 490 \text{ m s}^{-1}$$ A1

(iii) mean translational K.E., $\langle E_k \rangle = \frac{1}{2}m\langle c^2 \rangle = \frac{3}{2}kT$

hence, $\langle c^2 \rangle \propto T$ M1

gradient of the graph $\propto \frac{1}{\langle c^2 \rangle}$

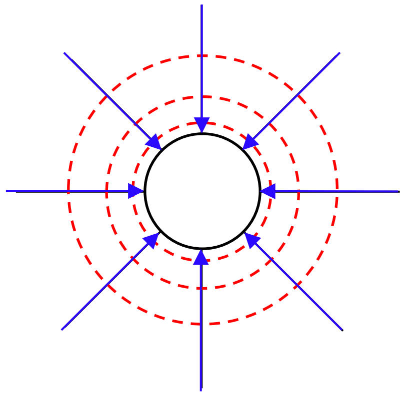
gradient of graph at T < gradient of graph at 290 K

$$\frac{1}{\langle c^2 \rangle_T} < \frac{1}{\langle c^2 \rangle_{290K}}$$

$$\langle c^2 \rangle_T > \langle c^2 \rangle_{290K}$$

$$T > 290 \text{ K}$$ A1

3 **(a)**



(i) *8 straight lines drawn with arrows directed radially towards the centre of the sphere. No electric field lines inside the sphere. B1

(ii) *3 concentric circles drawn around the sphere, distance between the lines increases with distance from the sphere. B1

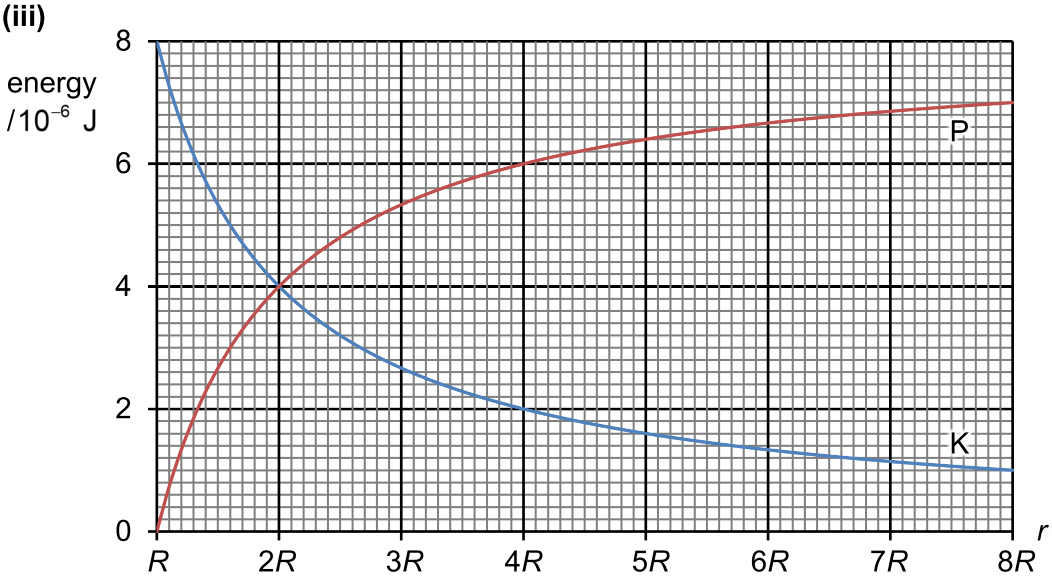
(b) **(i)** The electric force on the electron is the resultant force on the electron.

$$F_E = m_e a$$

$$a = \frac{F_E}{m_e} = \frac{eQ}{4\pi\varepsilon_0 R^2}\frac{1}{m_e} = \frac{eQ}{4\pi\varepsilon_0 m_e R^2}$$ B1

(ii) Since the acceleration of the electron is <u>inversely proportional to the square of the distance from the centre of the sphere</u>, the <u>speed</u> of the electron <u>increases at a decreasing rate</u> as it <u>moves radially away</u> from the centre of the sphere. B1 B1

(iii)



initial K.E. of electron at X $= 0$
final K.E. of electron $= 7.0 \times 10^{-6}$ J

initial E.P.E of electron at X, $P_X = \dfrac{1}{4\pi\varepsilon_0}\dfrac{(-e)(-Q)}{R} = \dfrac{1}{4\pi\varepsilon_0}\dfrac{eQ}{R}$

final E.P.E of electron at Y, $P_Y = \dfrac{1}{4\pi\varepsilon_0}\dfrac{(-e)(-Q)}{8R} = \dfrac{1}{8}\left(\dfrac{1}{4\pi\varepsilon_0}\dfrac{eQ}{R}\right) = \dfrac{1}{8}P_X$

decrease in E.P.E. $=$ increase in K.E.

$$P_X - \frac{1}{8}P_X = (7.0 - 0)\times 10^{-6}$$
$$P_X = 8.0 \times 10^{-6} \text{ J}$$

Hence, $P_Y = \dfrac{1}{8}P_X = 1.0 \times 10^{-6}$ J

By the conservation of energy, at all values of r,
total energy, K.E. $+$ E.P.E. $= 8.0 \times 10^{-6}$ J

1. *P is a decreasing function, always positive B1
*correct shape, passing through correct points at $r = R,\ 2R,\ 4R,\ 8R$ B1

2. *K is an increasing function, always positive B1
*correct shape, passing through correct points at $r = R,\ 2R,\ 4R,\ 8R$ B1

*If no labels at all – deduct 1 mk overall

4 **(a)** **(i)** 1.02 V — B1

When G_1 shows a null deflection, the potential difference across the $2.04\ \text{k}\Omega$ resistor is equal in magnitude to *E*.

(ii) current through cell A, $I_A = \dfrac{V_1}{R_1}$

$$= \frac{1.02}{2.04 \times 10^3} \quad \text{M1}$$

$$= 5.00 \times 10^{-4}\ \text{A}$$

power supplied by cell A, $P = I_A E_1$

$$= 5.00 \times 10^{-4} \times 2.00$$

$$= 1.00 \times 10^{-3}\ \text{W} \quad \text{A1}$$

*Final answer given to 3 or 4 s.f.
*If final answer given to less than 3 s.f. – deduct s.f. mk from paper

(b) current through cell B, $I_B = \dfrac{E}{R_2 + r}$

$$= \frac{1.02}{1.50 + 0.50} \quad \text{M1}$$

$$= 0.510\ \text{A}$$

new voltmeter reading $V_1 = E - I_B r$ OR $V_1 = I_B R_2$

$= 1.02 - (0.510)(0.50)$ OR $= (0.510)(1.50)$

$= 0.765\ \text{V}$ $\quad$ $= 0.765\ \text{V}$ — A1

*Final answer given to 2 or 3 s.f.

(c) current through PQ, I_A = current through $2.04\ \text{k}\Omega$ resistor

$$= \frac{V_1}{R_1}$$

$$= \frac{0.765}{2.04 \times 10^3} \quad \text{M1}$$

$$= 3.75 \times 10^{-4}\ \text{A}$$

p.d. across PQ, $V_{PQ} = I_A R_{PQ}$

$$= \left(3.75 \times 10^{-4}\right)(20.0) \quad \text{M1}$$

$$= 7.50 \times 10^{-3}\ \text{V}$$

$$\frac{E_{thermo}}{V_{PQ}} = \frac{L_{PJ}}{L_{PQ}}$$

$$E_{thermo} = \frac{0.400}{1.20} \times \left(7.50 \times 10^{-3}\right)$$

$$= 2.50 \times 10^{-3}\ \text{V} \quad \text{A1}$$

5 **(a)** A steady direct current results in a magnetic flux linkage in the coil that does not vary with time / is constant. B1

By Faraday's law, there is no induced e.m.f. and hence no induced current in the coil. B1

(b) Using an ammeter to measure a live current requires disconnecting the current-carrying wire to insert it into the ammeter to connect it in series, which may not be safe or practical. B1

However, a current clamp can measure the live current without any physical contact with the wire itself and without having to stop the current. B1

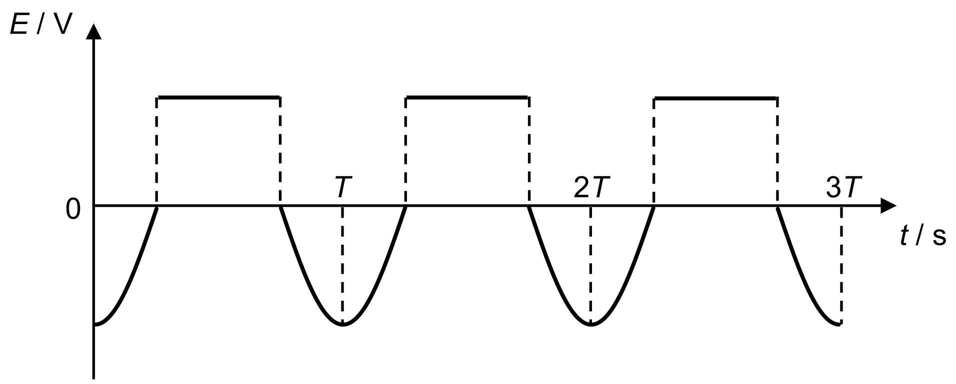
(c) **(i)** $\Phi = NBA$

$$B = \frac{\Phi}{NA}$$

$$= \frac{0.72 \times 10^{-6}}{9\left(2.0 \times 10^{-4}\right)}$$ M1

$$= 4.00 \times 10^{-4} \text{ T}$$ A1

(ii)



*correct sign of e.m.f. at each section within each period B1
*correct shape of graph at each section within each period B1
*If did not label horizontal axis – deduct 1 mk overall

(iii) Without the soft iron core to concentrate the magnetic field lines from the wire, the magnetic flux linkage and hence the rate of change of magnetic flux linkage of the coil would decrease. By Faraday's law, this decreases the induced e.m.f. M1

Hence the value of the r.m.s. current induced would decrease. A1

6 **(a)** **(i)**

$$E_{\text{photon}} = \frac{hc}{\lambda}$$
$$= \frac{\left(6.63\times10^{-34}\right)\left(3.00\times10^{8}\right)}{450\times10^{-9}}$$
$$= 4.42\times10^{-19}\ \text{J}$$ A1

(ii) 1.20 V is the stopping potential. B1

(iii) $E_{\text{photon}} = \Phi + eV_s$

$$\Phi = E_{\text{photon}} - eV_s$$
$$= 4.42\times10^{-19} - \left(1.60\times10^{-19}\right)(1.20)$$ M1
$$= 2.50\times10^{-19}\ \text{J}$$ A1

(iv) Increasing the intensity of light <u>increases the rate of incident photon but the energy of each photon remains unchanged</u>. B1

Since the light is incident on the same metal plate, <u>work function remains unchanged</u>. From Einstein's photoelectric equation, $E_{k,\max} = eV_s = E_{\text{photon}} - \Phi$, <u>maximum kinetic energy</u> of a photoelectron and hence <u>stopping potential remains unchanged</u>. B1

(b) **(i)** Since the electron in the atom is <u>bound to the nucleus, energy is required to remove this electron</u> to infinity, where potential energy is zero. Therefore, the energy levels are expressed with negative values. B1

(ii) Ionisation energy is the minimum amount of energy required for the outermost electron in the ground state to excite to the energy level of 0 eV to escape.
$$\Delta E = \left(0-(-10.4)\right)\times\left(1.60\times10^{-19}\right)$$
$$= 1.664\times10^{-18}$$
$$= 1.66\times10^{-18}\ \text{J}$$ B1

(iii)
$$\Delta E = \frac{hc}{\lambda}$$

$$\lambda = \frac{hc}{\Delta E}$$
$$= \frac{\left(6.63\times10^{-34}\right)\left(3.00\times10^{8}\right)}{(5.5-1.6)\left(1.60\times10^{-19}\right)}$$ M1
$$= 3.1875\times10^{-7} = 3.19\times10^{-7}\ \text{m}$$ A1

7 **(a)**

$$\lambda = \frac{\ln 2}{t_{1/2}}$$

$$= \frac{\ln 2}{87.7 \times 365 \times 24 \times 60 \times 60}$$ M1

$$= 2.5062 \times 10^{-10}\ \text{s}^{-1}$$

activity at launch,

$$A_0 = \lambda N_0$$

$$= \left(2.5062 \times 10^{-10}\right)\left(7.20 \times 10^{25}\right)$$ M1

$$= 1.8045 \times 10^{16} = 1.80 \times 10^{16}\ \text{Bq}$$ A1

(b) $P_0 = A_0 \times$ energy released per decay

$$= \left(1.80 \times 10^{16}\right) \times \left(5.50 \times 10^{6}\right)\left(1.60 \times 10^{-19}\right)$$ M1

$$= 15840 = 15800\ \text{W}$$ A1

(c) $P = A \times$ energy released per decay $= \lambda N \times$ energy released per decay

$\therefore P \propto N$

$$N = N_0 e^{-\lambda t}$$

$$\frac{N}{N_0} = \frac{P}{P_0} = e^{-(2.5062\times10^{-10})(10\times365\times24\times60\times60)} = 0.92401$$ M1

percentage of power available after 10 years is 92.4 %. A1

(d)
- Plutonium is relatively abundant hence cost-saving. Solar panels are expensive to build. B1
- Plutonium's long half-life ensures it can provide sufficient power to last the duration of the mission. Solar panels need to be directed towards the sun, which may not always be possible.
- As the distance between Sun and Saturn is large, to generate enough power, solar panels would have been too large and heavy to be practical.

(e) The difference in energy is in the <u>kinetic energy of the recoiling daughter nucleus</u> (uranium-234) which travels in the opposite direction to that of the α-particle due to conservation of linear momentum. B1

8 **(a)** Isotopes are nuclides that have the same number of protons but different number of neutrons. B1

(b) Bombard the gas atoms with high-energy electrons. B1
OR
Radiate the gas atoms with high-energy radiation / laser.

(c) The collimator allows only ions that are travelling perpendicularly to both the electric and magnetic fields to enter the Velocity Selector. B1

*Do not accept "allows ions to travel in a straight line" as such a path may not necessarily be normal to either or both fields.

(d) **(i)** increase in kinetic energy = decrease in electric potential energy

$$\frac{1}{2}mv^2 - 0 = q\Delta V \quad \text{M1}$$

$$v = \sqrt{\frac{2q\Delta V}{m}}$$

$$= \sqrt{\frac{2\left(1.60\times10^{-19}\right)\left(10.0\times10^{3}\right)}{\left(20\times1.66\times10^{-27}\right)}} \quad \text{M1}$$

$$= 310460$$

$$= 3.10\times10^{5}\ \text{m s}^{-1}\ \text{(shown)}$$

*If did not show non-rounded off value – deduct second M1 mark

(ii) Since magnetic force on the ions points to the left, the electric force must be of equal magnitude and points to the right.

$$F_E = F_B$$

$$qE = B_1qv$$

$$q\frac{\Delta V}{d} = B_1qv$$

$$\Delta V = B_1dv$$

$$= (0.160)(0.100)\left(3.10\times10^{5}\right) \quad \text{M1}$$

$$= 4960\ \text{V}$$

$$V = -4960\ \text{V} \quad \text{A1}$$

*Final answer to 3 or 4 s.f.
*Accept if use non-rounded off value of v and answer is -4960 V
*If V is positive – do not award A1

(iii) Since the speed of the ions is reduced, the magnetic force on the ions becomes smaller than the electric force. There is now a resultant force on the ions that points to the right. This causes the ions move / bend rightwards. M1 A1

Note: $qE = B_1qv$ is no longer true.

(iv) $B_2qv = m\frac{v^2}{r} \quad \Rightarrow \quad r = \frac{mv}{B_2q} \quad \Rightarrow \quad r \propto m$ B1

Draw a semicircular path with a larger radius than that of neon-20 ions since neon-21 ions have larger mass.

(e) relative atomic mass $= (20)(0.9051) + (21)(0.0027) + (22)(0.0922)$ M1

$= 20.1871$

$= 20.19$ (4 s.f.) A1

(f) When charge is doubled, $\dfrac{m}{z}$ is halved.

Draw 3 lines at $\dfrac{m}{z} =$ 10, 10.5 and 11. B1

*Ignore heights of lines.

(g) Change the <u>potential of the Ioniser to be negative</u> such that its potential is lower than the potential of the Collimator. This ensures that the negative ions are accelerated in the Accelerator. B1

<u>Reverse the direction of the magnetic field B_2</u> in the Ion Separator such that it points out of the page. This ensures that the magnetic force acting on the negative ions continues to point to the left as the ions enter the Ion Separator. B1

*Do not accept 'reverse directions of the electric and magnetic fields in the Velocity Selector'.